Metaheuristic Optimization Techniques on Benchmark Functions:

A Comparative Study Using GA, PSO, and Hill Climbing

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Introduction

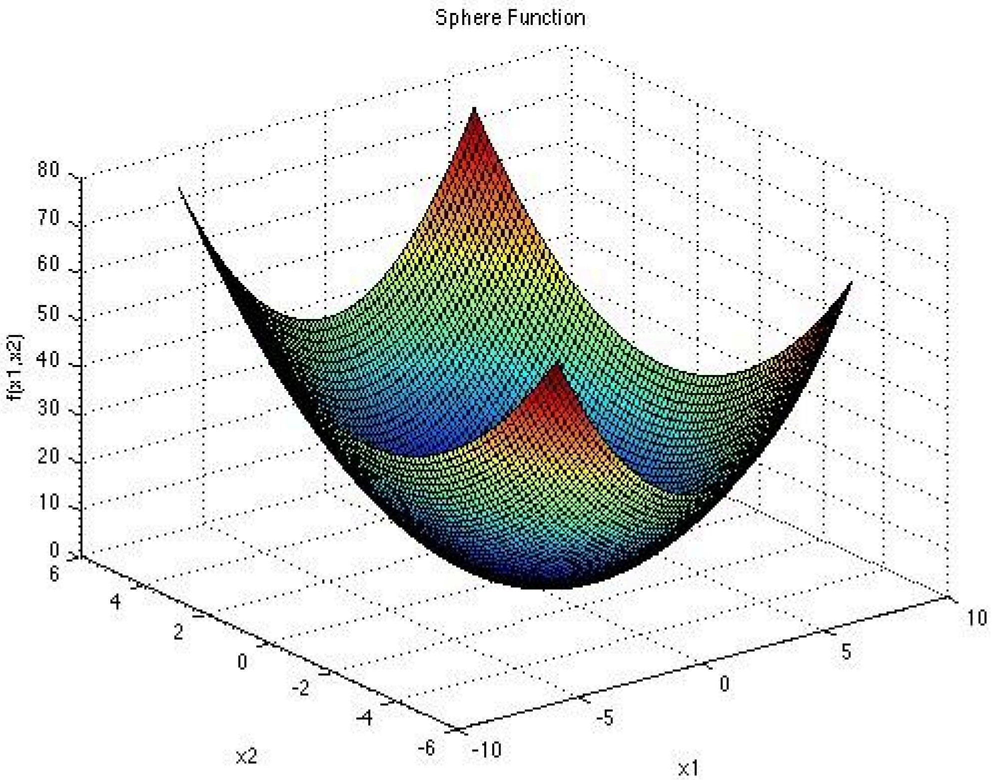
This project applies three optimization techniques: Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Hill Climbing (HC). These algorithms represent different optimization strategies, including evolutionary computation, swarm intelligence, and local search. The objective is to evaluate and compare their performance on benchmark functions.

Two benchmark functions, the Sphere function and the Rastrigin function, are used in this study. The performance of the algorithms is evaluated based on accuracy, convergence behavior, and consistency across multiple runs. The results provide insights into the strengths and limitations of each algorithm.

Benchmark Function Selection

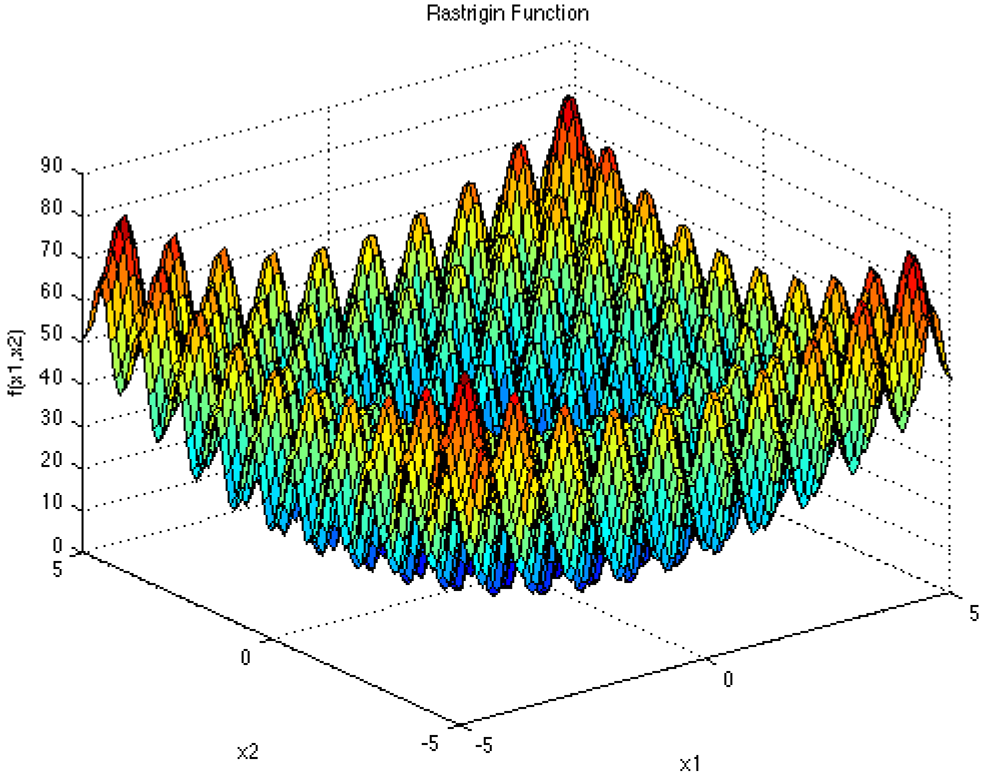
In this project, two widely used benchmark functions, the Sphere function and the Rastrigin function, are selected to evaluate the performance of the optimization algorithms. These functions are commonly used in optimization research because they represent different levels of complexity and provide a balanced evaluation environment.

The Sphere function is a simple unimodal function characterized by a smooth and convex search space. It contains only one global minimum and no local minima, making it an ideal function for testing the convergence speed and stability of optimization algorithms. Due to its simplicity, most algorithms can easily find the optimal solution. Therefore, the Sphere function is primarily used to evaluate how quickly and efficiently an algorithm converges toward the global optimum.



$$f(\mathbf{x}) = \sum_{i=1}^{d} x_i^2$$

Rastrigin function is a highly multimodal function with a large number of local minima distributed across the search space. This makes it significantly more challenging for optimization algorithms, as they may become trapped in local optima rather than reaching the global minimum. The oscillatory nature of the Rastrigin function creates a complex landscape, requiring strong exploration capabilities from the algorithms.



$$f(\mathbf{x}) = 10d + \sum_{i=1}^{d} \left[x_i^2 - 10\cos(2\pi x_i)\right]$$

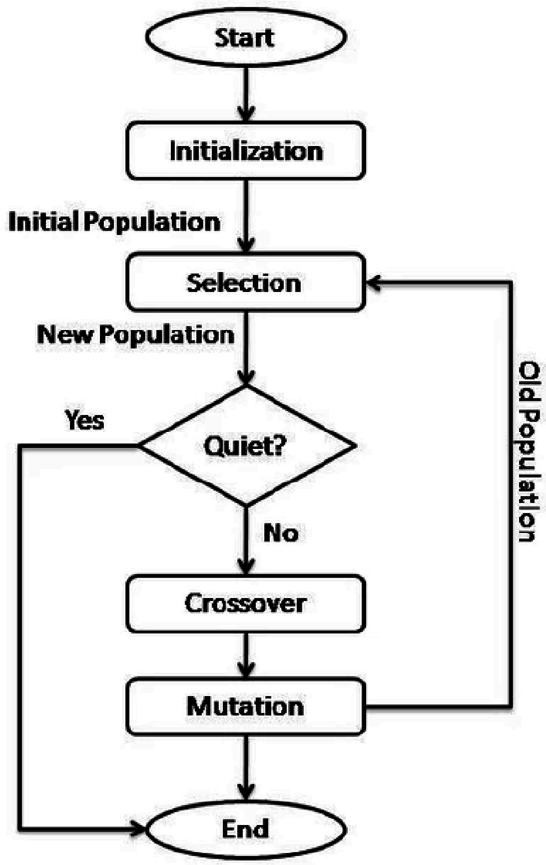
Both functions have their global minimum located at the origin, where the function value is zero. The typical search domain for both functions ranges from -5.12 to 5.12 in each dimension. By using these two benchmark functions, the study evaluates both the exploitation ability of the algorithms (through the Sphere function) and their exploration capability (through the Rastrigin function).

Figures included in this section illustrate the surface plots of both functions. The Sphere function displays a smooth bowl-shaped surface, while the Rastrigin function shows a highly irregular landscape with multiple peaks and valleys, highlighting the difference in optimization difficulty.

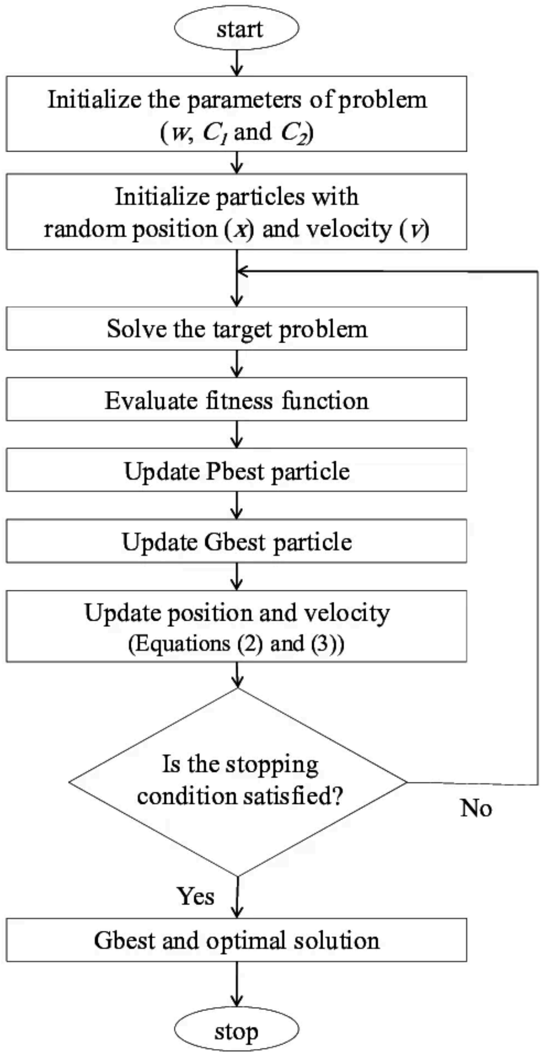
Algorithm Selection

In this study, three optimization algorithms are selected: Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Hill Climbing (HC). These algorithms represent different categories of optimization techniques, allowing a comprehensive comparison of their performance on both simple and complex benchmark functions.

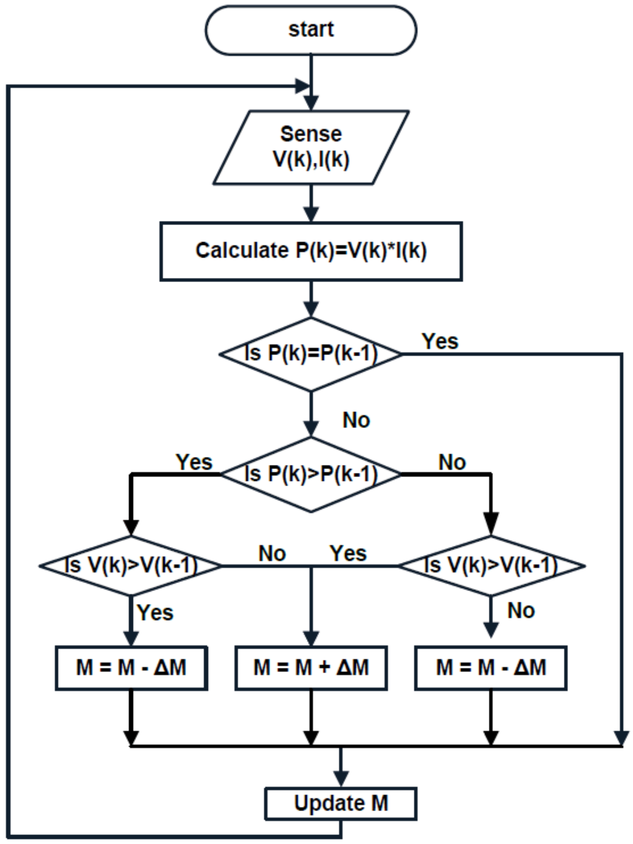
The **Genetic Algorithm** is an evolutionary approach inspired by natural selection. It evolves a population of candidate solutions using selection, crossover, and mutation, making it suitable for complex optimization problems with multiple local minima.



Particle Swarm Optimization is a swarm intelligence-based algorithm inspired by the social behavior of birds. It updates candidate solutions using both individual and global best positions, resulting in fast convergence.



Hill Climbing is a local search algorithm that iteratively improves a solution by moving to better neighboring solutions. Although efficient, it is prone to getting trapped in local optima.



These algorithms were selected to represent different optimization strategies: evolutionary (GA), swarm-based (PSO), and local search (Hill Climbing), enabling a comprehensive performance comparison.

Literature Review

Research on metaheuristic optimization algorithms highlights their effectiveness in solving complex optimization problems. Genetic Algorithm (GA) is widely used for its strong global search capability, while Particle Swarm Optimization (PSO) is known for fast convergence in continuous spaces. However, PSO may struggle with premature convergence in complex landscapes.

Simple methods such as Hill Climbing provide efficient local search but are limited by their inability to escape local optima. Despite extensive research on these algorithms, comparative analysis across different benchmark functions remains an important area of study.

Metrics such as fitness value and convergence behavior provide useful insights into algorithm performance. These considerations guide the approach of this project in evaluating GA, PSO, and HC on selected benchmark functions.

Methodology

The optimization process was implemented in Python using numerical and optimization libraries, including NumPy, Matplotlib, PySwarms, and DEAP. The study applies three algorithms: Particle Swarm Optimization (PSO), Genetic Algorithm (GA), and Hill Climbing (HC) to solve benchmark functions.

The benchmark functions define the search environment, where each algorithm iteratively updates candidate solutions to minimize the objective function. The Sphere function represents a smooth search space, while the Rastrigin function introduces a complex landscape with multiple local minima.

For PSO, a swarm of 30 particles was initialized in a 10-dimensional search space. The algorithm used an inertia weight of 0.9 with cognitive and social parameters set to 0.5 and 0.3, respectively. The particles updated their positions over 100 iterations based on individual and global best solutions.

For GA, a population of 50 individuals was used. The algorithm applied crossover with a probability of 0.7 and mutation with a probability of 0.2 over 100 generations. Selection and variation operators were used to evolve solutions toward optimal values.

Hill Climbing was implemented as a local search method, starting from a random solution and iteratively exploring neighboring solutions using a step size of 0.1. The algorithm was executed for up to 1000 iterations.

Each algorithm was evaluated over 30 independent runs to ensure consistency. The experiments generated performance data, including fitness values, convergence trends, and statistical summaries, enabling comparison of algorithm behavior across different problem landscapes.

Results and Analysis

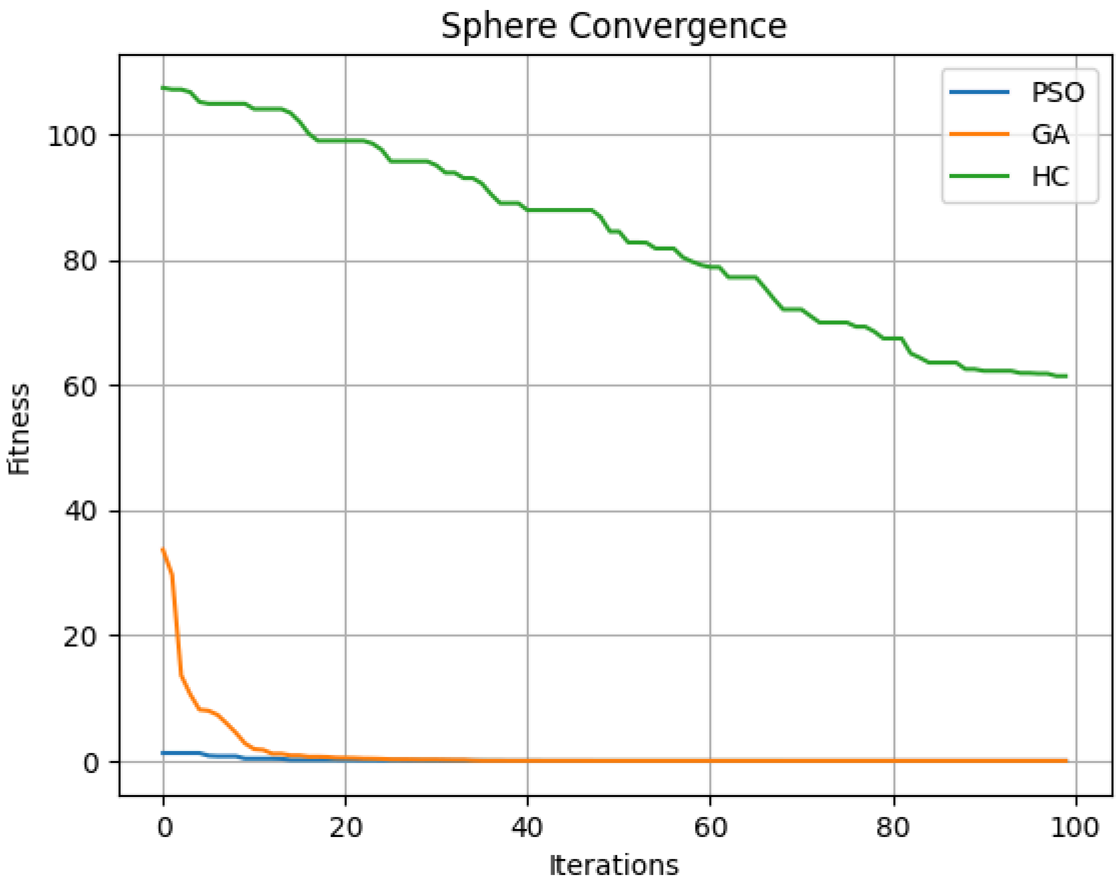
• Accuracy: PSO achieved near-optimal values on the Sphere function (≈0.001), while GA obtained slightly higher values (≈0.01). On the Rastrigin function, GA performed better (≈1.3) compared to PSO (≈2.5), while Hill Climbing (HC) produced higher values, indicating weaker performance.

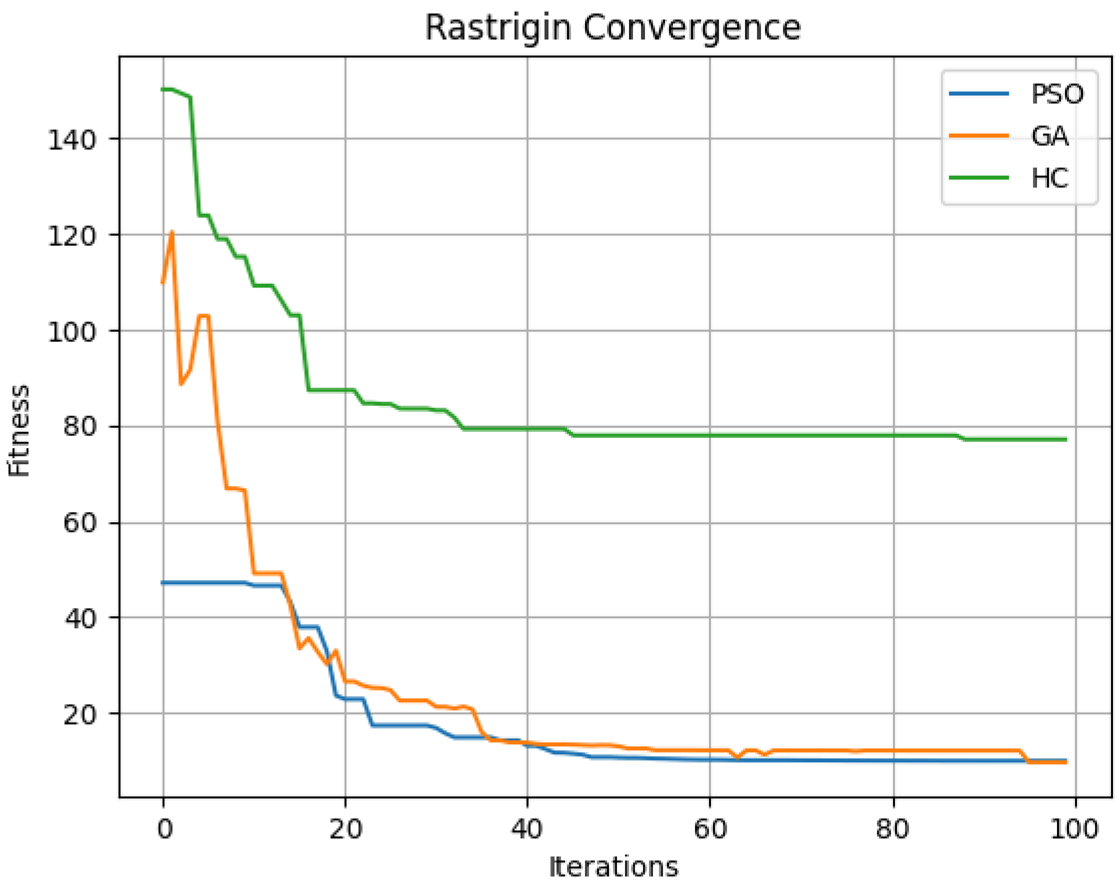
• Convergence: The convergence plots show that PSO converges faster on the Sphere function, reaching optimal values within fewer iterations. GA demonstrates a more gradual and stable improvement, particularly on the Rastrigin function. HC shows rapid initial improvement but stagnates early due to local minima.

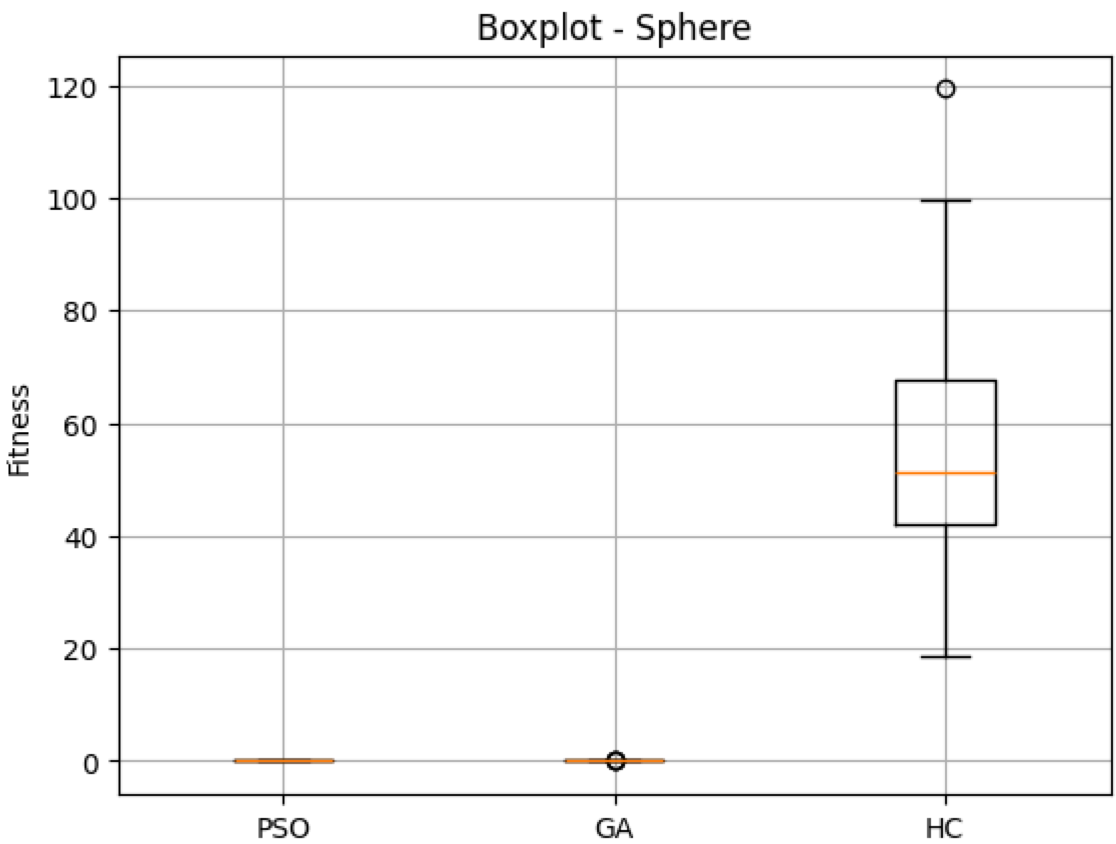
• Stability: Boxplot analysis across 30 runs shows that PSO and GA have more consistent performance with lower variability, while HC exhibits higher variation and less stability.

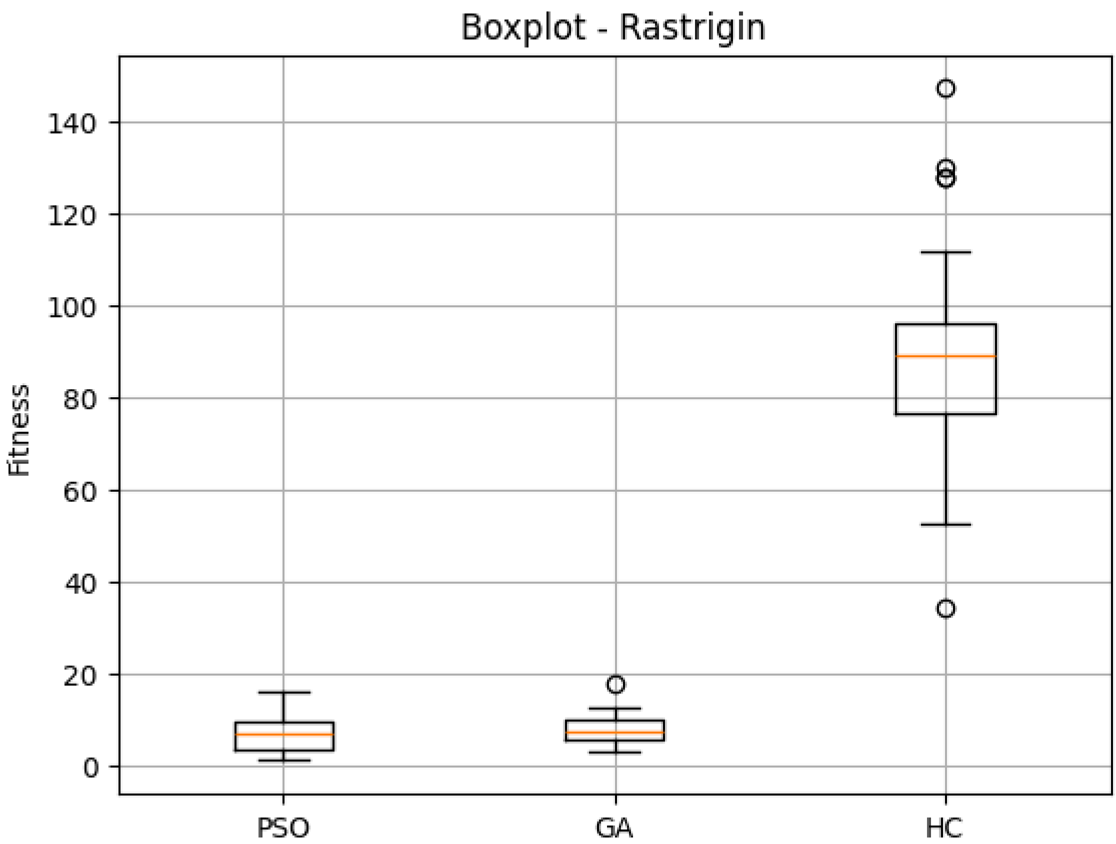
• Cost: HC required the least computational time due to its simplicity, followed by PSO, while GA required more time because of population-based operations.

Overall, the results indicate that PSO is efficient for simple optimization problems, while GA is more robust for complex multimodal functions. The convergence graphs highlight speed differences, and boxplots confirm stability and consistency across runs.




Rastrigin Convergence
PSO
GA
HC
Fitness
Iterations
0
20
40
60
80
100
120
140


Boxplot - Sphere
Fitness
0
20
40
60
80
100
120
PSO
GA
HC



Algorithm	Fitness (Sphere)	Fitness (Rastrigin)	Runtime (s)	Computational Cost
PSO	0.000039	6.678633	~0.012	Medium
GA	0.000946	7.689129	~0.133	High
HC	46.831079	90.336585	~0.001	Low

Conclusion

The optimization model reveals differences in algorithm performance across problem types, showing how each method responds to varying search landscapes. PSO demonstrates fast convergence and efficient performance on simple functions, while GA provides stronger exploration and better results on complex multimodal problems. Hill Climbing shows quick initial improvement but is limited by local optima.

The analysis highlights that performance metrics such as fitness and convergence provide valuable insights into algorithm behavior. Convergence plots illustrate differences in optimization speed, while boxplot analysis confirms variations in stability and consistency across multiple runs. These findings guide the selection of appropriate optimization techniques based on problem complexity.

Future work could explore hybrid approaches and adaptive parameters to further improve optimization performance.